

AFRILANG Stdlib

std/c/statbayes

Français. Bibliothèque AFRILANG 'std/c/statbayes' (niveau complex, catégorie complex). Elle expose 7 fonction(s) : bayesUpdate, posteriorOdds, priorFromOdds, oddsFromPrior, bayesFactor, credibleInterval, expectedLoss.

```
'import "std/c/statbayes"' · 'use statbayes'
```

- 'bayesUpdate(prior number, likelihood number, evidence number) → number'
- 'posteriorOdds(priorOdds number, likelihoodRatio number) → number'
- 'priorFromOdds(odds number) → number'
- 'oddsFromPrior(prior number) → number'
- 'bayesFactor(likelihoodH1 number, likelihoodH0 number) → number'
- 'credibleInterval(samples list number) → list number'

English. AFRILANG library 'std/c/statbayes' (tier complex, category complex). Exports 7 function(s): bayesUpdate, posteriorOdds, priorFromOdds, oddsFromPrior, bayesFactor, credibleInterval, expectedLoss.

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Catégorie / Category	Modules complex (c/)	
Niveau / Tier	complex	June 16, 2026
Fonctions / Functions	7	

Contents

Français

1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/statbayes' (niveau complex, catégorie complex). Elle expose 7 fonction(s) : bayesUpdate, posteriorOdds, priorFromOdds, oddsFromPrior, bayesFactor, credibleInterval, expectedLoss.

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- `bayesFactor(likelihoodH1 number, likelihoodH0 number) → number`
- `credibleInterval(samples list number) → list number`
- `expectedLoss(losses list number, probs list number) → number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/statbayes"
use statbayes
```

Niveau : complex · Catégorie : Modules complex (c/)
Domaines avancés – graphes, NLP, réseaux complexes.

3. Référence API

- bayesUpdate(prior number, likelihood number, evidence number) → number•
- posteriorOdds(priorOdds number, likelihoodRatio number) → number•
- priorFromOdds(odds number) → number•
- oddsFromPrior(prior number) → number•
- bayesFactor(likelihoodH1 number, likelihoodH0 number) → number•
- credibleInterval(samples list number) → list•
- expectedLoss(losses list number, probs list number) → number

4. Exemples d'utilisation

```
import "std/c/statbayes"
use statbayes
```

```
create result = bayesUpdate(/* args */)
say result
```

```
create result = posteriorOdds(/* args */)
say result
```

```
create result = priorFromOdds(/* args */)
say result
```

```
create result = oddsFromPrior(/* args */)
say result
```

```
create result = bayesFactor(/* args */)
say result
```

```
create result = credibleInterval(/* args */)
say result
```

5. Bonnes pratiques

- Préférez ``import "std/c/statbayes"`` en tête de fichier.
- Lancez ``afrilang check`` avant de livrer.
- Combinez avec les modules stdlib de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

6. Annexe — source

module statbayes

```
function bayesUpdate(prior number, likelihood number, evidence number) returns number
end
```

```
function posteriorOdds(priorOdds number, likelihoodRatio number) returns number
end
```

```
function priorFromOdds(odds number) returns number
end
```

```
function oddsFromPrior(prior number) returns number
end
```

```
function bayesFactor(likelihoodH1 number, likelihoodH0 number) returns number
end
```

```
function credibleInterval(samples list number) returns list number
end
```

```
function expectedLoss(losses list number, probs list number) returns number
end
end
```

English

1. Overview

AFRILANG library 'std/c/statbayes' (tier complex, category complex). Exports 7 function(s): bayesUpdate, posteriorOdds, priorFromOdds, oddsFromPrior, bayesFactor, credibleInterval, expectedLoss.

'import "std/c/statbayes"' · 'use statbayes'

```
- `bayesUpdate(prior number, likelihood number, evidence number) → number`
- `posteriorOdds(priorOdds number, likelihoodRatio number) → number`
- `priorFromOdds(odds number) → number`
- `oddsFromPrior(prior number) → number`
- `bayesFactor(likelihoodH1 number, likelihoodH0 number) → number`
- `credibleInterval(samples list number) → list number`
- `expectedLoss(losses list number, probs list number) → number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/statbayes"
use statbayes
```

Tier: complex · Category: Complex modules (c/)
Advanced domains – graphs, NLP, complex networks.

3. API reference

- bayesUpdate(prior number, likelihood number, evidence number) → number•
- posteriorOdds(priorOdds number, likelihoodRatio number) → number•
- priorFromOdds(odds number) → number•
- oddsFromPrior(prior number) → number•
- bayesFactor(likelihoodH1 number, likelihoodH0 number) → number•
- credibleInterval(samples list number) → list•
- expectedLoss(losses list number, probs list number) → number

4. Usage examples

```
import "std/c/statbayes"
use statbayes
```

```
create result = bayesUpdate(/* args */)
say result
```

```
create result = posteriorOdds(/* args */)
say result
```

```
create result = priorFromOdds(/* args */)
say result
```

```
create result = oddsFromPrior(/* args */)
say result
```

```
create result = bayesFactor(/* args */)
say result
```

```
create result = credibleInterval(/* args */)
say result
```

5. Best practices

- Prefer ``import "std/c/statbayes"`` at the top of your file.
- Run ``afrilang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

6. Appendix — source

module statbayes

```
function bayesUpdate(prior number, likelihood number, evidence number) returns number
end
```

```
function posteriorOdds(priorOdds number, likelihoodRatio number) returns number
end
```

```
function priorFromOdds(odds number) returns number
end
```

```
function oddsFromPrior(prior number) returns number
end
```

```
function bayesFactor(likelihoodH1 number, likelihoodH0 number) returns number
end
```

```
function credibleInterval(samples list number) returns list number
end
```

```
function expectedLoss(losses list number, probs list number) returns number
end
end
```

Référence rapide / Quick reference

- Import : `import "std/c/statbayes"`
- Vérification : `afrilang check`
- Documentation : <https://afrilang.dev/stdlib/std/c/statbayes/>

Source (excerpt)

```
module statbayes

function bayesUpdate(prior number, likelihood number, evidence number) returns number
end

function posteriorOdds(priorOdds number, likelihoodRatio number) returns number
end

function priorFromOdds(odds number) returns number
end

function oddsFromPrior(prior number) returns number
end

function bayesFactor(likelihoodH1 number, likelihoodH0 number) returns number
end

function credibleInterval(samples list number) returns list number
end

function expectedLoss(losses list number, probs list number) returns number
end
end
```